Iso Lab (EKS Spot, L2) replay of the priced DP/eq residual index-calculus study, p36

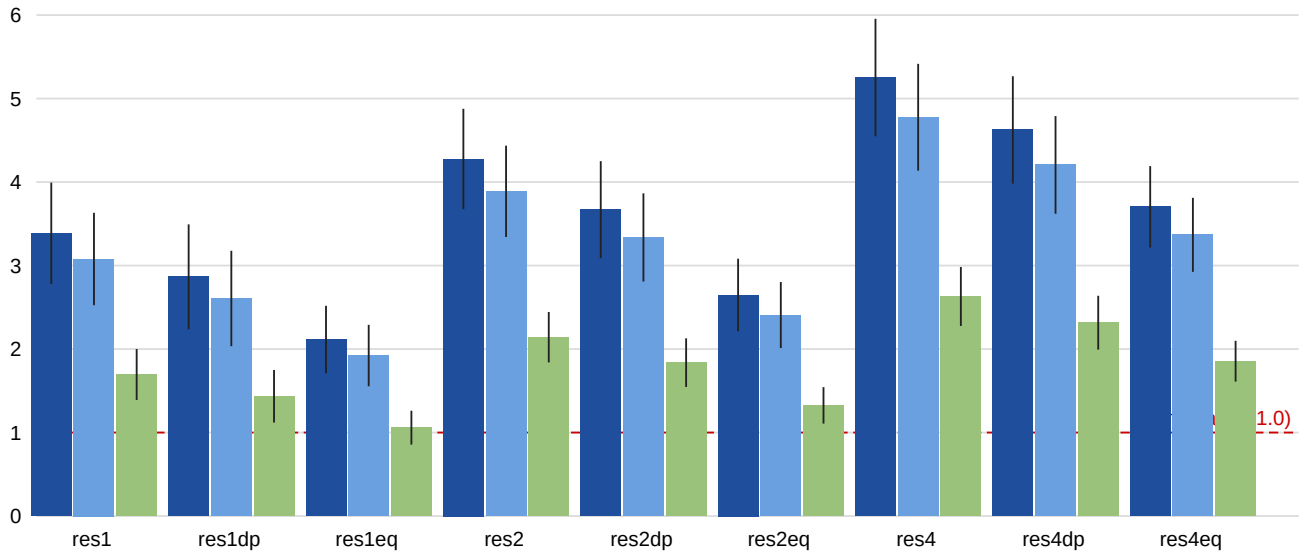
Class: infrastructure verification. Not a speedup. The goal (index calculus near rho parity) is not met: the best arm (res1eq) is still 2.11 × rho at w=0 on this estimator (the w>0 values, 1.92 and 1.06, price only rho's counters because the IC records carry none, so they understate the IC arms' cost and are not a band result; the harness band is authoritative).

Outcomes

- **R1 determinism.** All 528 records common to the local L0 run (rounds 1–3 × 16 workloads × 11 arms) have identical counted GAE, recovered scalar and algorithm seed (11 arms × 48 = 528/528).
- **R2 admission gates.** Session complete; audit.json ok, 0 problems, 1056 records; 880 replays, all `identical`; L2 gate accepted all 1056 executions (80/80 pairs at L2); A/A res2 vs res2aa wall ratio 0.9995 [0.9981, 1.0017] (within [0.9, 1.1]); quality score 85. SHA256SUMS: 19/19 OK. Source commit be464efac844aa20598a769a085820d2c7101ae9 (+ musl ioctl fix), ecbench sha256 2ce2d100…ed95, image digest sha256:dcb51a3a…83e0bb, Job UID 4c53b06b-5201-47f7-9dc2-116df3697d21.
- **Harness comparison res2aa/rho** (priced w=0/0.1/1): 3.2491 / 3.2489 / 3.2478, CI95 [2.918, 3.617] over 80 pairs on 16 workloads. Wall ns/GAE ratio 1.4629 [1.442, 1.484], outside the A/A noise: the IC arm costs ~46% more wall time per counted operation than rho (memory-bound lookups). It is an upper bound on cost per operation, not a verdict.

Ratios to matched rho (my recomputation from the 5-round isolab records)

Mean over 16 workloads of the per-workload mean of per-round paired ratios total_gae(arm)/total_gae(rho), plus w × the uncharged counters where the record carries them. Columns: Iso Lab (5 rounds) ± 95% t half-width, then the local L0 (3 rounds) value. This estimator differs slightly from the harness's (3.34 vs 3.249 for res2aa at w=0); the harness comparison is authoritative, this table shows the arm ordering. Note the uncharged counters are in the record only for rho in this estimator, so w>0 columns are approximations; the harness band (above) is authoritative.



bars: price w=0 (dark), 0.1 (mid), 1 (light); cost ratio to matched rho, mean of per-workload paired ratios; whiskers = 95% t interval over 16 workl
Iso Lab EKS Spot, L2, 5 rounds, p36 (r=68719022539). Lower is better.

arm	w=0 isolab	L0	w=0.1 isolab	L0	w=1 isolab	L0
res1	3.385 ± 0.608	3.150	3.079 ± 0.553	2.864	1.695 ± 0.305	1.577
res1dp	2.865 ± 0.628	2.751	2.606 ± 0.571	2.502	1.435 ± 0.315	1.377
res1eq	2.114 ± 0.404	2.047	1.922 ± 0.368	1.862	1.059 ± 0.203	1.025
res2	4.277 ± 0.601	3.947	3.889 ± 0.547	3.589	2.142 ± 0.302	1.976
res2dp	3.669 ± 0.581	3.320	3.337 ± 0.528	3.019	1.837 ± 0.291	1.662
res2eq	2.647 ± 0.435	2.560	2.407 ± 0.395	2.328	1.326 ± 0.218	1.282

res4	5.252 ± 0.703	4.887	4.776 ± 0.640	4.444	2.630 ± 0.353	2.447
res4dp	4.624 ± 0.643	4.387	4.205 ± 0.585	3.989	2.316 ± 0.322	2.196
res4eq	3.704 ± 0.487	3.426	3.368 ± 0.443	3.115	1.855 ± 0.245	1.715

Caveats

- Same unpriced residual-table-bytes counter caveat as the parent study (validator F1); owner decision outstanding.
- Pre-existing validation gate: no benchmarks/README row until VALIDATION.json covers artifact-store raw data.
- One curve (p36); rounds 4–5 are new data with no L0 counterpart.

Evidence: s3 launcher receipt and artifacts of run 41342132360; protocol research/ic_residual_dp_eq_isolab_20261101/PROTOCOL.md (Amendments 1–2).